6 EXAM PRACTICE

1 The enthalpy changes of formation of gaseous ethene and gaseous ethane are 52 kJ mol^{-1} and –85 kJ mol^{-1} respectively at 298 K.

What is the enthalpy change of reaction at 298 K for the following process?

$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$

A −137 kJ mol^{-1} **B** −33 kJ mol^{-1}

C +33 kJ mol^{-1} **D** +137 kJ mol^{-1} [1]

(Total for Question 1 = 1 mark)

2 The enthalpy change for the neutralisation reaction below is −114 kJ mol^{-1}.

$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

Use this information to suggest the most likely value for the enthalpy change for the following neutralisation reaction.

$Ba(OH)_2(aq) + 2HCl(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

A −57 kJ mol^{-1} **B** −76 kJ mol^{-1}

C −114 kJ mol^{-1} **D** −228 kJ mol^{-1} [1]

(Total for Question 2 = 1 mark)

3 Which of the following processes is endothermic?

A the condensation of steam

B the freezing of water

C the decomposition of calcium carbonate into calcium oxide and carbon dioxide

D the reaction between hydrogen ions and hydroxide ions [1]

(Total for Question 3 = 1 mark)

4 For which of the following reactions does the value of $\Delta H^{\ominus}$ represent both a standard enthalpy change of formation and a standard enthalpy change of combustion?

A $C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$ **B** $C(g) + \frac{1}{2}O_2(g) \rightarrow CO(g)$

C $2C(s) + O_2(g) \rightarrow 2CO(g)$ **D** $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$ [1]

(Total for Question 4 = 1 mark)

5 The value of the enthalpy change for the process represented by the equation

$K(s) \rightarrow K^+(g) + e^-$

is equal to

A the electron affinity of potassium

B the enthalpy change of vaporisation of potassium

C the sum of the enthalpy change of atomisation and the first ionisation energy of potassium

D the sum of the enthalpy change of atomisation and the electron affinity of sodium [1]

(Total for Question 5 = 1 mark)

6 Which equation represents the change corresponding to the enthalpy change of atomisation of iodine?

A $I_2(g) \rightarrow 2I(g)$ **B** $I_2(l) \rightarrow 2I(g)$

C $I_2(s) \rightarrow 2I(g)$ **D** $\frac{1}{2}I_2(s) \rightarrow I(g)$ [1]

(Total for Question 6 = 1 mark)

7 Lead forms several solid oxides, the most common of which are PbO, PbO_2 and Pb_3O_4.

This question is about the enthalpy changes that take place during reactions involving these oxides.

(a) (i) State what is meant by enthalpy change of formation. [1]

(ii) State the standard conditions of temperature and pressure that are usually used in calculations involving enthalpy changes. [2]

(iii) Write an equation representing the standard enthalpy change of formation of PbO(s). [2]

(b) 'Red lead' (Pb_3O_4) can be made by heating PbO in oxygen:

$3PbO(s) + \frac{1}{2}O_2(g) \rightarrow Pb_3O_4(s)$

Calculate the standard enthalpy change for this reaction. [3]

$\Delta_f H^{\ominus}$ PbO(s) = −219 kJ mol^{-1}

$\Delta_f H^{\ominus}$ Pb_3O_4(s) = −735 kJ mol^{-1}

(c) When red lead is heated it decomposes into PbO and PbO_2.

$Pb_3O_4(s) \rightarrow 2PbO(s) + PbO_2(s)$

Use this information, together with the data supplied in part (b), to calculate the standard enthalpy change of formation of PbO_2(s). [3]

(Total for Question 7 = 11 marks)

8 Propane, C_3H_8, is a gas at room temperature. It is used as a fuel for portable gas cookers.

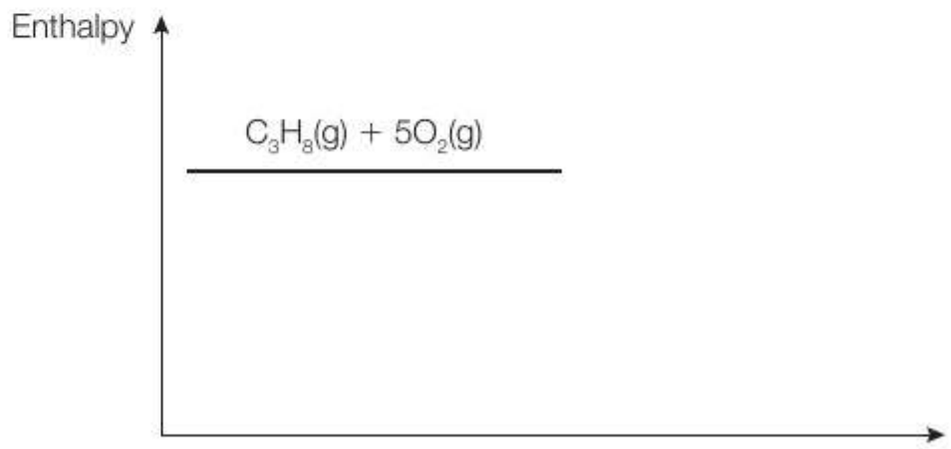
(a) Give two properties of propane that make it suitable for use as a fuel. [2]

(b) The standard enthalpy change of combustion of propane is represented by the equation:

$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$ $\Delta_c H^{\ominus} = 2220$ kJ mol^{-1}

(i) State what is meant by standard enthalpy change of combustion. [3]

(ii) Complete the enthalpy level diagram for the combustion of propane. Label $\Delta_c H^{\ominus}$. [2]



(c) Hess's Law can be used to calculate enthalpy changes of formation from enthalpy changes of combustion.

The equation for the formation of propane is

$3C(s) + 4H_2(g) \rightarrow C_3H_8(g)$

(i) State Hess's law. [2]

(ii) The standard enthalpy changes of combustion of carbon and hydrogen are $-394\,kJ\,mol^{-1}$ and $-286\,kJ\,mol^{-1}$ respectively. Calculate the standard enthalpy change of formation of propane gas. [2]

(Total for Question 8 = 11 marks)

9 Bond enthalpies can be used to calculate enthalpy changes of reaction.

Some bond enthalpies are given in the table.

Bond	Bond enthalpy / kJ mol^{-1}
H – H	+436
F – F	+158
H – Cl	+431
H – F	+562

(a) (i) State what is meant by bond enthalpy. [2]

(ii) State why the sign for a bond enthalpy is always positive. [1]

(b) The enthalpy change of formation of hydrogen chloride is represented by the following equation:

$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightarrow HCl(g) \qquad \Delta_f H^{\ominus} = -92\,kJ\,mol^{-1}$

Use this information, and the data in the table, to calculate the bond enthalpy of the Cl–Cl bond. [3]

(c) Use the data in the table to calculate the enthalpy change of formation of HF(g). [2]

(Total for Question 9 = 8 marks)

10 The enthalpy change of neutralisation for the reaction between hydrochloric acid, HCl(aq), and sodium hydroxide, NaOH(aq), can be determined using the following method:

- Place 50.0 cm^3 of 2.00 mol dm^{-3} HCl(aq) into a polystyrene cup and measure its temperature.
- Place 50.0 cm^3 of 2.10 mol dm^{-3} NaOH(aq) into another polystyrene cup and measure its temperature.
- Mix the two solutions, stir with the thermometer and record the highest temperature reached.

The results of one experiment were:

- Initial temperature of both HCl(aq) and NaOH(aq) = 20 °C
- Highest temperature reached by the mixture = 33.6 °C

The equation for the reaction is:

$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$

(a) State what is meant by enthalpy change of neutralisation. [2]

(b) Give the name of a piece of apparatus that is suitable for measuring the volumes of acid and alkali. [1]

(c) State why the NaOH(aq) used had a slightly higher concentration than the HCl(aq). [1]

(d) (i) Calculate the heat energy transferred, using the equation:

$Q = mc\Delta T$ [2]

[Assume the specific heat capacity of the final solution is $4.18\,J\,g^{-1}K^{-1}$ and that its density is $1\,g\,cm^{-3}$.]

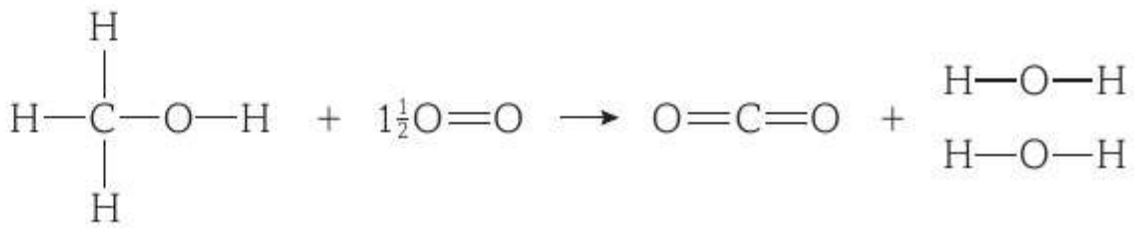
(ii) Calculate the enthalpy change of neutralisation for the reaction. [3]

(Total for Question 10 = 9 marks)

11 The table gives the values of some mean bond enthalpies.

Bond	Bond enthalpy / kJ mol^{-1}
C – H	+412
O – H	+463
O = O	+496
C = C	+743
C – O	+360

The equation for the combustion of methanol (CH_3OH) in the gaseous state is:



(a) (i) Use the data in the table to calculate the enthalpy change of combustion of gaseous methanol. [2]

(ii) Give two reasons why the standard enthalpy change of combustion of methanol is different from the value calculated in part (a)(i). [2]

(b) Which process measures the mean bond enthalpy for the C–H bond in methane? [1]

A $CH_4(g) \rightarrow CH_3(g) + H(g)$
ΔH = A; mean bond enthalpy = A

B $CH_4(g) \rightarrow C(g) + 4H(g)$
ΔH = B; mean bond enthalpy = B/4

C $CH_4(g) \rightarrow C(g) + 2H_2(g)$
ΔH = C; mean bond enthalpy = C/4

D $CH(g) \rightarrow C(g) + H(g)$
ΔH = D; mean bond enthalpy = D

(c) Calculate the mean bond enthalpy of the S–F bond in SF_6 given the following data. [2]

$SF_6(g) \rightarrow S(s) + 3F_2(g) \qquad \Delta H = +1100\,kJ\,mol^{-1}$

$S(s) \rightarrow S(g) \qquad \Delta H = +223\,kJ\,mol^{-1}$

$F_2(g) \rightarrow 2F(g) \qquad \Delta H = +158\,kJ\,mol^{-1}$

(Total for Question 11 = 7 marks)